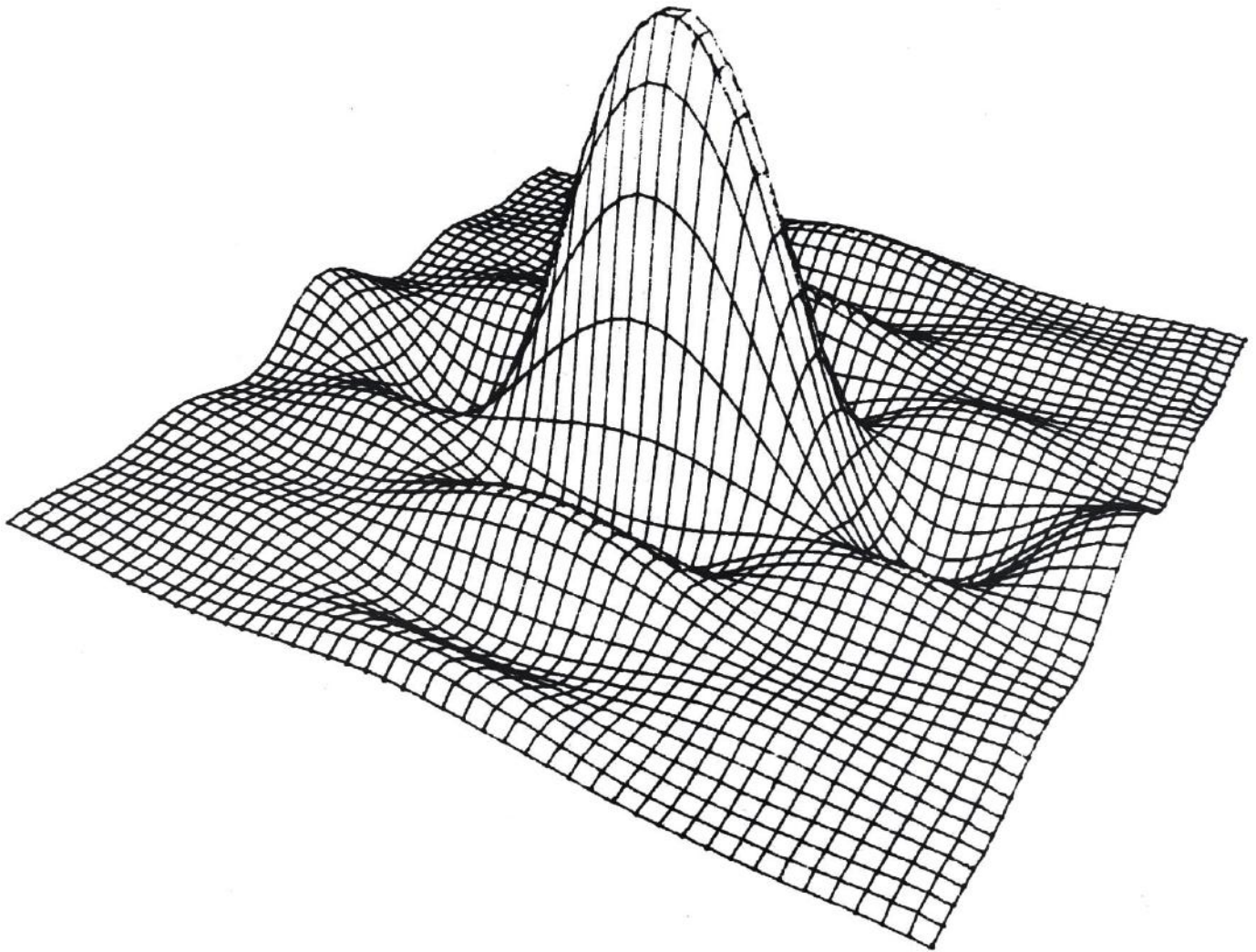


MATH SURVEY



STUYVESANT H.S.
SPRING 1979

We dedicate this issue of Math Survey to
seventy-five years of quality education at
Stuyvesant. We are particularly proud of
the Math Department, renowned throughout
the country for its excellence.

MATH SURVEY

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Spring 1979 Volume LIV

TABLE OF CONTENTS

The Uses of Fallacy	1 - 4
$2 + 2 \stackrel{?}{=} 0$	5
Math Society	6
Area of an n-gon	7 - 8
Westinghouse Tips	9
Diced Probability	10
Regular Convex Polyhedra	11 - 12
The Five Color Theorem	13 - 14
Matrix Equations	15 - 16
Stereoscopic 3-D Plots	Centerfold Insert
Cryptography	19 - 20
Triangular and Square Numbers	21 - 22
Wallis's Product	23
Math Team	24
Extinction Probability	25 - 26
Cardinality of Polynomial Functions	27 - 28
Stirling's Approximation	29 - 32
N-th Powers of Integers	33 - 35

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Uses of Fallacy

In the last hundred years or so, mathematics has undergone a tremendous growth in size and complexity and subtlety. This growth has given rise to a demand for more flexible methods of proving theorems than the laborious, difficult, pedantic, 'rigorous' methods previously in favor. This demand has been met by what is now a well-developed branch of mathematics known as Generalized Logic. I don't want to develop the theory of Generalized Logic in detail, but I must introduce some necessary terms. In Classical Logic, a Theorem consists of a True Statement for which there exists a Generalized Proof. I think that the meaning of these terms should be sufficiently clear without the need for elaborate definitions.

The applications of Generalized Proofs will be obvious. Professional authors of text-books use them freely, especially when proving mathematical results in Physics texts. Teachers and lecturers find that the use of Generalized Proofs enables them to make complex ideas readily accessible to students at an elementary level (without the necessity for the tutor to understand them himself). Research workers in a hurry to claim priority for a new result, or who lack the time and inclination to be pedantic, find Generalized Proofs useful in writing papers. In this application, Generalized Proofs have the further advantage that the result is not required to be true, thus eliminating a tiresome (and now superfluous) restriction on the growth of mathematics.

I want now to consider some of the proof techniques which Generalized Logic has made available. I will be concerned mostly with the ways in which these methods can be applied in lecture courses - they require only trivial modifications to be used in text-books and research papers.

The reductio methods are particularly worthy of note. There are, as everyone knows, two reductio methods available: reductio ad nauseam and reductio ad erratum. Both methods begin in the same way: the mathematician denies the result he is trying to prove, and writes down all the consequences of this denial that he can think of. The methods are most effective if these consequences are written down at random, preferably in odd vacant corners of the blackboard.

Although the methods begin in the same way, their aims are completely different. In reductio ad nauseam the lecturer's aim is to get everyone in the class asleep and not taking notes. (The latter is a much stronger condition.) The lecturer then has only to clean the blackboard and announce, 'Thus we arrive at a contradiction, and the result is established.' There is no need to shout this - it is the signal for which everyone's subconscious has

been waiting. The entire class will awaken, stretch, and decide to get the last part of the proof from someone else. If everyone had stopped taking notes, therefore, there is no 'someone else', and the result is established.

In reductio ad erratum the aim is more subtle. If the working is complicated and pointless enough, an error is bound to occur. The first few such mistakes may well be picked up by an attentive class, but sooner or later one will get through. For a while, this error will lie dormant, buried deep in the working, but eventually it will come to the surface and announce its presence by contradicting something which has gone before. The theorem is then proved.

It should be noted that in reductio ad erratum the lecturer need not be aware of this random error or of the use he has made of it. The best practitioners of this method can produce deep and subtle errors within two or three lines and surface them within minutes, all by an instinctive process of which they are never aware. The subconscious artistry displayed by a really virtuoso master to a connoisseur who knows what to look for can be breathtaking.

There is a whole class of methods which can be applied when a lecturer can get from his premises P to a statement A , and from another statement B to the desired conclusion C , but he cannot bridge the gap from A to B . A number of techniques are available to the aggressive lecturer in this emergency. He can write down A , and without any hesitation put 'therefore B '. If the theorem is dull enough, it is unlikely that anyone will question the 'therefore'. This is the method of Proof by Omission, and is remarkably easy to get away with (sorry, 'remarkably easy to apply with success').

Alternatively, there is the Proof by Misdirection, where some statement that looks rather like ' A , therefore B ' is proved. A good bet is to prove the converse ' B , therefore A ': this will always satisfy a first-year class. The Proof by Misdirection has a countably infinite analogue, if the lecturer is not pressed for time, in the method of Proof by Convergent Irrelevancies.

Proof by Definition can sometimes be used: the lecturer defines a set S of whatever entities he is considering for which B is true, and announces that in the future he will be concerned only with members of S . Even an Honors class will probably take this at face value, without inquiring whether the set S might not be empty.

Proof by Assertion is unanswerable. If some vague waffle about why B is true does not satisfy the class, the lecturer simply says, 'This point should be intuitively obvious. I've explained it as clearly as I can. If you still cannot see it, you will just have to think very carefully about it yourselves, and then you will see how trivial and obvious it is.'

The hallmark of a Proof by Admission of Ignorance is the

statement, 'None of the textbooks makes this point clear. The result is certainly true, but I don't know why. We will just have to accept it as it stands.' This otherwise satisfactory method has the potential disadvantage that somebody in the class may know why the result is true (or, worse, know why it is false) and be prepared to say so.

A Proof by Non-Existent Reference will silence all but the most determined troublemaker. 'You will find a proof of this given in Copson on page 445', which is in the middle of the index. An important variant of this technique can be used by lecturers in pairs. Dr. Jones assumes a result which Professor Smith will be proving later in the year - but Professor Smith, finding himself short of time, omits that theorem, since the class has already done it with Dr. Jones....

Proof by Physical Reasoning provides uniqueness theorems for many difficult systems of differential equations, but it has other important applications besides. The cosine formula for a triangle, for example, can be obtained by considering the equilibrium of a mechanical system. (Physicists then reverse the procedure, obtaining the conditions for equilibrium of the system from the cosine rule rather than from experiment.)

The ultimate and irrefutable standby, of course, is the self-explanatory technique of Proof by Assignment. In a text-book, this can be recognized by the typical expressions 'It can readily be shown that...' or 'We leave as a trivial exercise for the reader the proof that...'. (The words 'readily' and 'trivial' are an essential part of the technique.)

An obvious and fruitful ploy when confronted with the difficult problem of showing that B follows from A is the Delayed Lemma. 'We assert as a lemma, the proof of which we postpone...'. This is by no means idle procrastination: there are two possible denouements. In the first place, the lemma may actually be proved later on, using the original theorem in the argument. This proof by Circular Cross-Reference has an obvious inductive generalization to chains of three or more theorems, and some very elegant results arise when this chain of interdependent theorems becomes infinite.

The other possible fate of a Delayed Lemma is the Proof by Infinite Neglect, in which the lecture course terminates before the lemma has been proved. The lemma, and the theorem of which it is a part, will naturally be assumed without comment in future courses.

A very subtle method of proving a theorem is the method of Proof by Osmosis. Here the theorem is never stated, and no hint of its proof is given, but by the end of the course it is tacitly assumed to be known. The theorem floats about in the air during the entire course and the mechanism by which the class absorbs it is the well-known biological phenomenon of osmosis.

A method of proof which is regrettably little used in under-

graduate mathematics is the Proof by Aesthetics ('This result is too beautiful to be false'). Physicists will be aware that Dirac uses method to establish the validity of several of his theories, the evidence for which is otherwise fairly slender. His remark 'It is more important to have beauty in one's equations than to have them fit experiment' has achieved a certain fame.

I want to discuss finally the Proof by Oral Tradition. This method gives rise to the celebrated Folk Theorems, of which Fermat's Last Theorem is an imperfect example. The classical type exists only as a footnote in a text-book, to the effect that it can be proved (see unpublished lecture notes of the late Professor Green) that. . . . Reference to the late Professor Green's lecture notes reveals that he had never actually seen the proof, but had been assured of its validity in a personal communication, since destroyed, from the great Sir Ernest White. If one could still track it back from here, one would find that Sir Ernest White heard of it over coffee one morning from one of his research students, who had seen a proof of the result, in Swedish, in the first issue of a mathematical magazine which never produced a second issue and is not available in the libraries. And so on. Not very suprisingly, it is common for the contents of a Folk Theorem to change dramatically as its history is investigated.

I have made no mention of Special Methods such as division by zero, taking wrong square roots, manipulating divergent series, and so forth. These methods, while very powerful, are adequately described in the standard literature. Nor have I discussed the little-known Fundamental Theorem of All Mathematics, which states that every number is zero (and whose proof will give the interested reader hours of enjoyment, and excellent practice in the use of the methods outlined above). However, it will have become apparent what riches there are in the study of Generalized Logic, and I appeal to Mathematics Departments to institute formal courses in this discipline. This should be done preferably at undergraduate level, so that those who go teaching with only a Bachelor's degree should be familiar with the subject. It is certain that in the future nobody will be able to claim a mathematical education without a firm grounding in at least the practical applications of Generalized Logic.

Note from the editors:

The above was adapted from an article
published by P. V. Dunmore in the New
Zealand Mathematics Magazine 1970, Vol. 7, 15

2+2=0

by Gregg Patruno

Theorem: All numbers equal zero

Proof: Let $S = 1 - 1/2 + 1/3 - 1/4 + 1/5 - 1/6 + \dots$

Since S is the sum of the positive quantities $(1 - 1/2)$, $(1/3 - 1/4)$, $(1/5 - 1/6)$, ..., $S > 0$.

$(1 - S)$ is also the sum of the positive quantities, i.e. $(1/2 - 1/3) + (1/4 - 1/5) + (1/6 - 1/7) + \dots$, so $1 - S > 0$, or $S < 1$. Therefore S converges to a particular value.

$$\begin{aligned}
 2S &= 2(1 - 1/2 + 1/3 - 1/4 + 1/5 - 1/6 + \dots) \\
 &= 2(1 + 1/3 + 1/5 + \dots) - 2(1/2 + 1/4 + 1/6 + \dots) \\
 &= 2(1 + 1/3 + 1/5 + \dots) - (1 + 1/2 + 1/3 + \dots) \\
 &= 2(1 + 1/3 + 1/5 + \dots) - (1 + 1/3 + 1/5 + \dots) \\
 &\quad + (1/2 + 1/4 + 1/6 + \dots) \\
 &= 2(1 + 1/3 + 1/5 + \dots) - (1 + 1/3 + 1/5 + \dots) \\
 &\quad - (1/2 + 1/4 + 1/6 + \dots) \\
 &= (1 + 1/3 + 1/5 + \dots) - (1/2 + 1/4 + 1/6 + \dots) \\
 &= 1 - 1/2 + 1/3 - 1/4 + 1/5 - 1/6 + \dots = S
 \end{aligned}$$

$2S = S$. Since $S \neq 0$, we can divide by S , obtaining $2 = 1$. Subtracting 1 from both sides, $1 = 0$.

Multiplying by any number x , $x = 0$.

Therefore all numbers are equal to zero. Q.E.D.

Alternate proof:

Consider the equation $x^{x^{x^{\dots}}} = 2$. The exponent, which is again $x^{x^{x^{\dots}}}$, is also 2, and we have $x^2 = 2$. Therefore, $x = \sqrt{2}$, so $\sqrt{2}^{\sqrt{2}} = 2$.

Now consider the equation $x^{x^{x^{\dots}}} = 4$. The exponent is also

4, so $x^4 = 4$ and x again equals $\sqrt{2}$. Therefore,

$$\sqrt{2}\sqrt{2}\sqrt{2}\cdots = 4.$$

But we have already determined that

$$\sqrt{2}\sqrt{2}\sqrt{2}\cdots = 2, \text{ so } 2 \text{ must equal } 4.$$

$$2 = 4$$

$$0 = 2 \quad (\text{subtracting } 2)$$

$$0 = x \quad (\text{multiplying by } x/2)$$

Therefore, $x = 0$ for all x .

Q.E.D.

Math Society

Math Society's function is to bring higher mathematics to the students of Stuyvesant. It has been very successful this year, presenting informative and varied lectures.

We have been fortunate in obtaining outstanding speakers such as Professor Raymond Smullyan, who introduced us to recreational logic. A lecture on Euclidean Geometry given by Professor Posimentier was especially well received. Other topics discussed this year included cyclic numbers, probability, topology, non-Euclidean Geometry, graph theory, computer logic (Turing machines), and catastrophe theory, a new and undeveloped field of mathematics. In addition to the previously mentioned speakers, we were honored to have with us Prof. David C. Kelly.

Math Society president Steven Lev, vice-president Aaron Jungreis, and faculty advisor Stuart Weinberg were responsible for organizing this year's club activities. We expect Math Society to continue to meet with great success and we encourage all students with an interest in mathematics to attend. Lectures are given in room 207 after 3:00. Announcements are posted outside the math office and elsewhere a few days prior to each lecture, and refreshments are served during the informal discussion at the end of the lecture.

Westinghouse Tips

While we realize that Nucleus has already published an article on the Westinghouse, we feel that since our staff won a total of \$12,500 to their 0, we may be somewhat more qualified to give advice on that contest.

Of course, it is impossible to issue a set of fixed rules which will guarantee success. There is no sure way of winning. You can't just decide to do a Westinghouse project which will win, nor will entering in conformance to someone else's wishes or joining a projects/research class make you a winner. Bronx Science, which has a projects class and which puts heavy pressure on its students to enter contests had more entrants than Stuyvesant, more honors winners (though less proportionally), slightly more finalists (far less proportionally), and no top prize winners at all.

On the other hand, there is no doubt that if you are already interested in a project, there are ways to improve your chances of winning. First, don't get discouraged. There is a temptation to think that your work is not significant, that most of it has been done before, or that your results are not satisfactory. (I not only went through all of these stages but had several university professors confirm the worthlessness of my work.) Just patch up the work and keep going. Of course, on the topic of repeating previously done work, research your project extensively before doing anything else. Public libraries are unlikely to be of much help except for providing general background information. However, by talking to the right people at universities it may be possible to gain access to their libraries. Professors are usually very eager to help really interested students, and may well be able to give advice on where to look for material. Nevertheless, there is no doubt that researching is the most time consuming and least interesting aspect of any project.

While the content of the work is obviously important, it may not be the decisive factor. The work should be attractively and seriously presented (all but one or two Westinghouse judges were born with long faces), and well documented. Don't stretch your work out (winning papers have been as short as 2 or 3 pages) but don't pay too much attention to the length limitation either.

Finally, work hard on the data blank, (known to entrants as the DATA BLANK). It is inane, pretentious, and nearly impossible to complete, but generally recognized as an important factor in the judging. Spend time on your section of the form, and choose a teacher who will do the same on the teacher's section.

In short, recognize that even Westinghouse judges are people, too. Their technical knowledge is sufficient for them to judge the content of virtually any work (they include specialists in various fields), but they are also affected by surface impressions. So, do the best work you can and leave yourself sufficient time to present it well. Best of luck!

Diced Probability

by Paul Feldman

If a die is rolled, the number obtained added to an increasing sum starting at zero, and the event repeated ad infinitum, what is the probability that at some point a particular integer is exactly equal to the sum?

For very small numbers, it is practical to count the ways the dice can add up to the given number. It can be shown that for $n = 1$ through 6, the probability of getting n is: $(7/6)^n/6$. This follows from the expansion of $(1 + 1/6)^n$. Of course, this will not work for larger numbers, or probabilities greater than one would be reached.

Let us say that we are at the point where the probability of hitting n , $n+1$, $n+2$, etc. are all the same. Assume we don't hit the number n . Let

k be the probability that $n-1$ was the last number hit before n ,
 m be the probability that $n-2$ was the last number hit before n ,
 p be the probability that $n-3$ was the last number hit before n ,
 q be the probability that $n-4$ was the last number hit before n ,
 r be the probability that $n-5$ was the last number hit before n .

It should be clear that for any series of rolls, exactly one of these events will happen. Also, if we keep rolling until the sum is about $2n$, we can count the rolls in reverse order. Thus, the same number k is the probability of $n-1$ being the first number hit after n ; however, this is really saying that k is the probability of $n+1$ being the first hit after n as well as the probability of $n-1$ being the last hit before n . The same symmetry argument shows that m , p , q , and r also apply to hitting $n+2$, $n+3$, $n+4$, and $n+5$, respectively, directly after n .

How could the event associated with r occur? Since n , $n+1$, $n+2$, $n+3$, and $n+4$ aren't hit, the sum before was $n-1$ (with probability k) after which a six was rolled (with probability $1/5$), since the conditions on the event associated with k imply that a 1 would not be rolled next. Thus $r = k/5$. For $n+4$ to be first hit after n , either $n-1$ was last hit and a five rolled, or $n-2$ was last hit and a six rolled. (Remember, after $n-2$ only 3, 4, 5, or 6 could be rolled.)

Thus, $q = 1/5 k + 1/4 m$

Similarly, $p = 1/5 k + 1/4 m + 1/3 p$ (Ways to hit $n+3$ first after n)

$n = 1/5 k + 1/4 m + 1/3 p + 1/2 q$

$k = 1/5 k + 1/4 m + 1/3 p + 1/2 q + r$

Now, to solve the five simultaneous equations in five variables, use $k = k/5 + m/4 + p/3 + q/2 + r$ and substitute $r = k/5$, so $k = k/5 + m/4 + p/3 + q/2 + k/5$, so $3k/5 = m/4 + p/3 + q/2$.
 $q = k/5 + m/4$, so $3k/5 = m/4 + p/3 + (1/5)(k/2) + (1/4)(m/2)$

continued on page 35

Regular Convex Polyhedra

by Gideon Eisenstadter

The regular solids have been known since ancient times. The three simplest - the tetrahedron, the cube, and the octahedron - were known to the Egyptians. The remaining two - the dodecahedron and the icosahedron - were discovered by Pythagoras. The curious fact that there are only five such solids (as will be proved later) intrigued the ancients. Plato discussed their supposed natural significance. The four elements have their characteristic shapes: the cube is assigned to the earth, the octahedron to air, the tetrahedron to fire, and the icosahedron to water. The fifth, the dodecahedron, was used by the Creator for the universe itself.

In later years, Kepler associated the five regular solids with the spaces between the planetary orbits; thus it was evident that there existed only six planets (only six were known at that time). Kepler represented the Earth's orbit by a sphere around which he circumscribed first a dodecahedron and then another sphere for the orbit of Mars. Further circumscription of a tetrahedron and a cube specified the orbits of Jupiter and Saturn, respectively. Within the sphere of Earth, he repeated this procedure with an icosahedron and an octahedron to determine the orbits of Mercury and Venus.

Theorem: There are exactly five regular polyhedra: the tetrahedron, the cube, the octahedron, the dodecahedron, and the icosahedron.

Proof: For a given polyhedron, let m denote the number of edges meeting at each vertex, and b the number of edges bounding each face. Furthermore, let e , v , and f denote the numbers of edges, vertices, and faces of the polyhedron.

Each of the faces is bounded by b edges and each edge belongs to two faces; m edges meet at each of the v vertices and each edge connects two vertices. Hence,

$$e = \frac{vm}{2} = \frac{fb}{2}$$

Solving for v and f ,

$$v = \frac{2e}{m}, \quad f = \frac{2e}{b} \tag{1}$$

Substituting in Euler's equation, $v + f - e = 2$, and dividing by e ,

$$\frac{2}{m} + \frac{2}{b} - 1 = \frac{2}{e} \tag{2}$$

11 In a polyhedron, $m \geq 3$ and $b \geq 3$. This implies that $m \leq 5$,

for if $m = 6$ and $b = 3$, then

$$\frac{2}{m} + \frac{2}{b} - 1 = \frac{1}{3} + \frac{2}{3} - 1 = 0,$$

which contradicts equation (2). Analogously, $b = 5$.

If $m = 4$ or $m = 5$, then $b = 3$. This also entails a proof by contradiction. Suppose that $m = 4$ and $b = 4$. Then

$$\frac{2}{m} + \frac{2}{b} - 1 = \frac{1}{2} + \frac{1}{2} - 1 = 0,$$

which again contradicts (2). Analogously, if $b = 4$ or 5 , then $m = 3$.

Thus there are five possible cases:

(i) $m = 3$, $b = 3$. Substituting in (2),

$$\frac{2}{e} = \frac{2}{3} + \frac{2}{3} - 1 = \frac{1}{3}, \quad e = 6.$$

From (1): $v = \frac{2e}{m} = \frac{12}{3} = 4$, $f = \frac{2e}{b} = \frac{12}{3} = 4$.

This describes a tetrahedron.

The four remaining cases result from similar substitutions into equations (2) and (1).

(ii) $m = 4$, $b = 3$. We get $e = 12$, $v = 6$, $f = 8$ -an octahedron.

(iii) $m = 3$, $b = 4$. We get $e = 12$, $v = 8$, $f = 6$ -a cube.

(iv) $m = 5$, $b = 3$. We get $e = 30$, $v = 12$, $f = 20$ -an icosahedron.

(v) $m = 3$, $b = 5$. We get $e = 30$, $v = 20$, $f = 12$ -a dodecahedron.

At this point we have shown that there are exactly five different types of polyhedra in which equal numbers of edges bound each face and equal numbers of edges meet at each vertex. Hence, there are at most five regular convex polyhedra. But since we can produce five, these must be the only cases. Q.E.D.

PROBLEMS

Sum and Simplify:

$$\frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \cdots + \frac{1}{n(n+1)} + \cdots + \frac{1}{9900}.$$

Isosceles triangle AEB, with base $\overline{AB}$, is drawn exterior to the square ABCD. If the area of EDC is 20 more than the area of AEB, find the area of the square.

The Five Color Theorem

by Joel Wein

What is the minimum number of colors needed to color a map so that no two adjacent faces have the same color? This is an old problem of mathematics. In 1977, it was proven that four is the minimum number (by use of a computer which checked nineteen hundred thirty-six topologically unequivalent cases). I cannot prove this, but I will prove that at most five colors are needed.

To do this, consider a map. On each face of the map, put a vertex. If two faces are adjacent, put an edge between their vertices. I will prove that this graph is planar, i.e. can be drawn with no two edges crossing each other (except at a vertex), we can five color the vertices.

Lemma: If a graph G is planar, then G has a vertex of degree less than or equal to five (i.e. a vertex with five or fewer adjacent edges).

Proof of Lemma: To do this it is necessary to show that in a planar graph, $3F/2 \leq E \leq 3V - 6$, where F is the number of faces, V is the number of vertices, and E is the number of edges. Case 1 is when the graph G has a face bounded by fewer than three edges. This implies that G is the graph in diagram 1, and the theorem holds ($3/2 \leq 2 \leq 3$).

Otherwise,

$3 \leq$ the number of edges bounding face 1.

$3 \leq$ the number of edges bounding face 2.

$\vdots$

$3 \leq$ the number of edges bounding face F .

Adding the columns, and calling the sum of the second column D , we get $3F \leq D$. Since no edge can border two faces, $D \leq 2E$, therefore $3F \leq 2E$, and $F \leq 2E/3$. Add $V - E$ to both sides to get $V - E + F \leq V - E/3$. By Euler's formula, $V - E + F = 2$, so $2 \leq V - E/3$, and $E \leq 3V - 6$. Therefore, $3F/2 \leq E \leq 3V - 6$.

Now let's assume our lemma is false. This means that there are no vertices with degree less than or equal to five.

Therefore,

$6 \leq$ the degree of vertex 1.

$6 \leq$ the degree of vertex 2.

$\vdots$

$6 \leq$ the degree of vertex v .

Since every edge is adjacent to two vertices, the sum of the second column is $2E$; therefore we have $6V \leq 2E$, or $3V \leq E$. This contradicts $E \leq 3V - 6$, which means the original assumption is wrong; therefore there exists a vertex of degree less than or equal to five.

Now we are ready for the proof of the five color theorem. I will do this by an induction on the number of vertices (V). The initial case, $V=1$, is obviously true because only one color is needed.

Assuming the theorem holds for all graphs with $V \leq K$, we must show that all graphs of $K+1$ vertices are five-colorable. By our lemma, we are guaranteed that there is a vertex of degree less than or equal to 5, which we will call A . Delete this vertex and all its adjacent edges, and call the new graph $G-A$. Since $G-A$ has K vertices by our hypothesis it can be colored in five colors. We must replace A and show we can color it with one of the five colors.

If there are less than five vertices adjacent to A , or there are five vertices adjacent to A but two are colored the same, we are guaranteed a fifth color for A . If not, we must recolor one of the five adjacent vertices.

Say the graph is like Diagram 1. Let's recolor P to 3. Then color all the vertices adjacent to P which were colored B to 1. Recolor all the vertices adjacent to these vertices 3. Continue this process as long as necessary. If we have not colored R 1, then we can give A color 1, and we are finished. But suppose the situation is something like Diagram 2: the coloring of P forces the coloring of R 1, and there is still no free color for A . To solve the problem, we can color Q 4, color all the adjacent vertices which were colored 4 2, as we did before. For this to force the coloring of S as 2, there would have to be a path of vertices colored only 2 and 4 going from Q to S , this path must intersect the path $PXMNJR$ somewhere. Since our graph is planar, any intersection must be at one of the vertices. But all of these vertices are colored 1 or 3. Therefore there can be no path of vertices colored 2 or 4 from Q to S , and we can color Q 4, give A color 2, and we have 5-colored our graph.

Diagram 1:

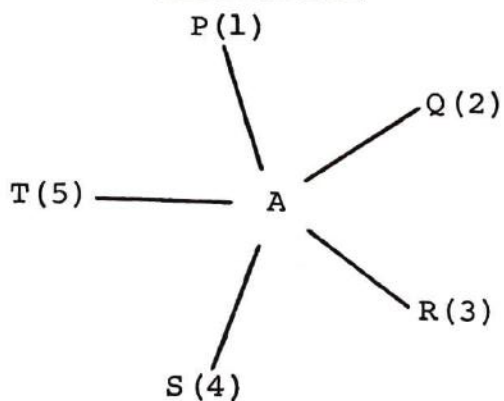
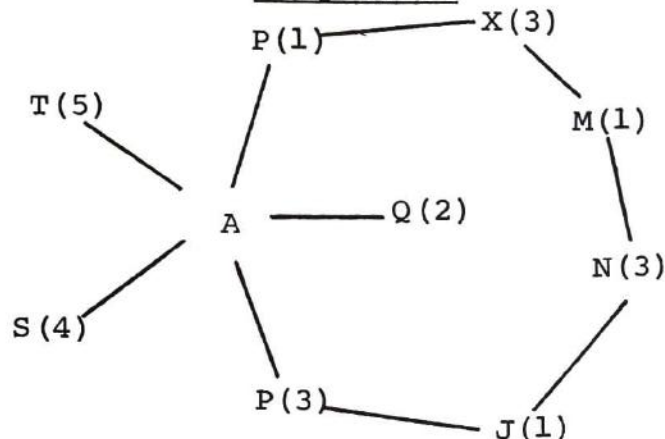


Diagram 2:



Matrix Equations

by Zachary Franco

How does one solve a quadratic equation in M_2 , the ring of 2×2 matrices with all of its entries belonging to C ? First we must find a method of taking the (or one of the) square root(s) of a matrix.

If A is a 2×2 matrix, say $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, we take an arbitrary matrix $B = \begin{pmatrix} x & y \\ z & w \end{pmatrix}$, and solve $B^2 = A$, or

$$(1) \quad x^2 + yz = a \qquad (3) \quad zx + wz = c$$

$$(2) \quad xy + yw = b \qquad (4) \quad yz + w^2 = d$$

If $b \neq 0$, then we may solve for x, w , and z in terms of y and substitute into (2), the other relations being satisfied, and we obtain:

$$(5) \quad [(a-d)y^2 - b^2]^2 = 4b^2y^2(d - (c/b)y^2), \text{ a quadratic in } y^2.$$

Hence, if $b \neq 0$, a 2×2 matrix can have at most four square roots which can be obtained by solving for y in (5) and obtaining x, z , and w .

If $b = 0$, and $c \neq 0$, we obtain $\begin{pmatrix} a & 0 \\ c & d \end{pmatrix}^{\frac{1}{2}} = \begin{pmatrix} \sqrt{a} & 0 \\ \frac{c}{\sqrt{a}+\sqrt{d}} & \sqrt{d} \end{pmatrix}$, where the signs of the radicals are not restricted (this also yields four square roots unless $a = d$, where we have two square roots).

If $b = 0$, and $c = 0$, $a \neq d$, we obtain:

$$\begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix}^{\frac{1}{2}} = \begin{pmatrix} \sqrt{a} & 0 \\ 0 & \sqrt{d} \end{pmatrix} \text{ with no restrictions on the signs.}$$

The only other cases left are the following two special cases:

(i) The square root of the identity matrix, $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ equals $\pm I$ or $\begin{pmatrix} r & s \\ \frac{1-r^2}{s} & -r \end{pmatrix}$, for any $r, s \in C, s \neq 0$. This result is

equivalent to the square root of I obtained from the Pauli identity: $(A + B + C)^2 = (\alpha^2 + \beta^2 + \gamma^2)I$, where

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, B = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, C = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

and α, β, γ can be complex.

The square root of $\begin{pmatrix} n & 0 \\ 0 & n \end{pmatrix}$ can be obtained by noting that

$$\begin{pmatrix} n & 0 \\ 0 & n \end{pmatrix}^{\frac{1}{2}} = n^{\frac{1}{2}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}^{\frac{1}{2}}$$

(ii) The square root of the zero matrix $0 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ equals zero or $\begin{pmatrix} r & s \\ -\frac{r}{s} & -r \end{pmatrix}$ for any $r, s \in \mathbb{C}, s \neq 0$.

Now we are ready to solve for the infinitely many matrices which satisfy: $x^2 + ax + bI = 0$. This equation is equivalent to:

$$(x + \frac{1}{2}aI)^2 = (\frac{1}{4}a^2I) - bI \quad \text{or}$$

$x = -\frac{1}{2}aI + \sqrt{\frac{1}{4}a^2 - b} \sqrt{I}$, where the radical sign denotes any square root. Similarly, the solutions to $x^2 + ax + M = 0$ are

$$-\frac{1}{2}aI + \sqrt{\frac{1}{4}a^2I - M} \sqrt{I} \text{ where } M \text{ is any matrix.}$$

There are many functions of matrices (from M to M) such as $\exp(A)$, $\ln(A)$, $\sin(A)$, etc. which have interesting properties. The easiest way to compute these matrices is by use of power series. (In fact, they are defined that way.) e.g.:

if $P = \begin{pmatrix} 0 & t \\ -t & 0 \end{pmatrix}$, note that $P^n = \begin{pmatrix} 0 & t^n \\ -t^n & 0 \end{pmatrix}, \begin{pmatrix} -t^n & 0 \\ 0 & -t^n \end{pmatrix}, \begin{pmatrix} 0 & -t^n \\ t^n & 0 \end{pmatrix}$, or

$\begin{pmatrix} t^n & 0 \\ 0 & t^n \end{pmatrix}$, depending on whether $n \equiv 1, 2, 3, \text{ or } 4 \pmod{4}$.

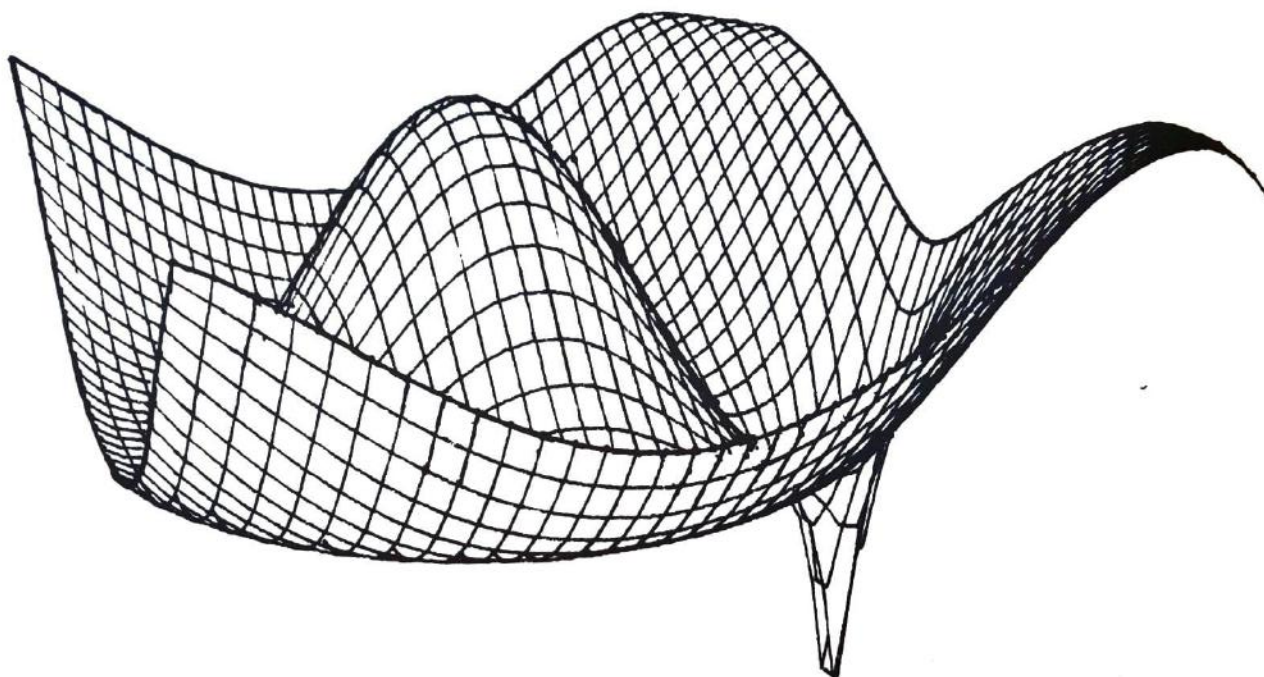
$$\text{So } \exp(P) = I + \sum_{k=1}^{\infty} \frac{1}{k!} P^k =$$

$$\begin{pmatrix} \sum_{k=0}^{\infty} \frac{t^{2k}}{(2k)!} (-1)^k & \sum_{k=0}^{\infty} \frac{t^{2k+1}}{(2k+1)!} (-1)^k \\ \sum_{k=0}^{\infty} \frac{t^{2k+1}}{(2k+1)!} (-1)^k & \sum_{k=0}^{\infty} \frac{t^{2k}}{(2k)!} (-1)^k \end{pmatrix} = \begin{pmatrix} \cos(t) & \sin(t) \\ -\sin(t) & \cos(t) \end{pmatrix}$$

Similarly, $\exp \begin{pmatrix} 0 & t \\ t & 0 \end{pmatrix} = \begin{pmatrix} \cosh(t) & \sinh(t) \\ \sinh(t) & \cosh(t) \end{pmatrix}$. It would be an

interesting exercise for the reader to find when e^A diverges.

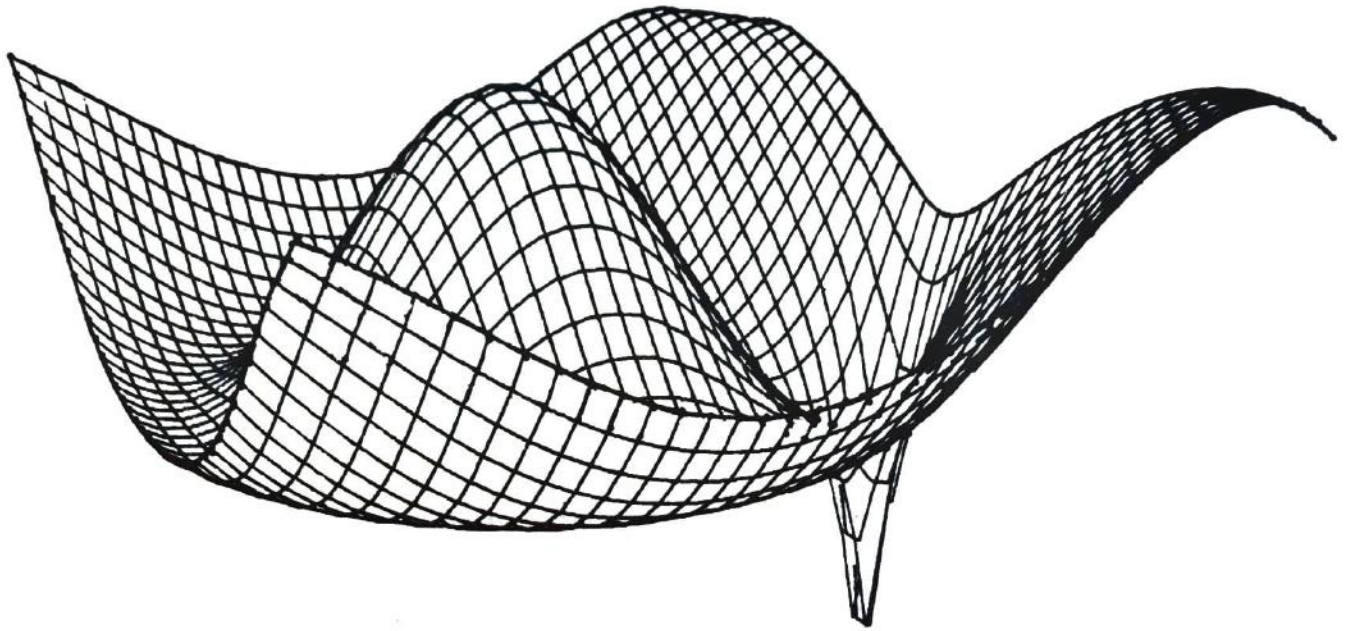
In concluding, I would like to demonstrate an interesting property of matrices. In solving $A^2 - 18A + 17I = 0$ we may factor to obtain: $(A - I)(A - 17I) = 0$ and we are tempted to say that either $A = I$ or $A = 17I$, but this reasoning does not work since $XY = 0$ does not imply that $X = 0$ or $Y = 0$.



Right eye.

INSTRUCTIONS

The pictures above form a stereoscopic image pair. Although at first glance they may look the same, in reality they are "snapshots" of a 3-D plot. (See "3-Dimensional Plotting," Math Survey, Spring 1977, pp. 6-8.) They are taken from two slightly different perspectives. If you view the plot on the left with your right eye and that on the right with your left eye by crossing your eyes, the function $z = \cos|x+iy| - 1.5/ (.3+|x+iy-4.5|)$ should jump out at you in 3-D. It is easier if you clear away surrounding objects and cover this text with a blank sheet. Borrowing eyeglasses from your farsighted friend or using a stereoscopic viewer will also help.



Left eye.

A BRAIN TEASER

The statement that the truth of either of the two statements that if hens don't lay eggs then frogs are insects or that if frogs are insects then all mice have bones implies that some mice don't have bones and also implies that either hens lay eggs or frogs are insects, implies that if either hens lay eggs or mice have bones then frogs are not insects. If frogs are insects then it is not true that if hens lay eggs then mice have bones.

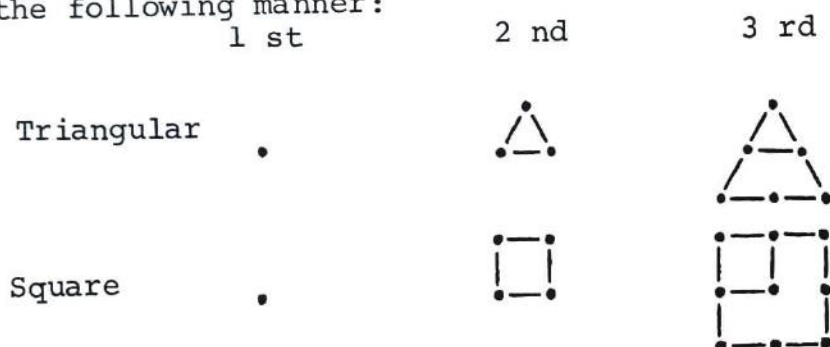
The above paragraph is equivalent to which of the following:

- 1) If hens don't lay eggs then either frogs are not insects or all mice have bones.
- 2) All mice have bones and if hens lay eggs then frogs are not insects.
- 3) Frogs are insects.
- 4) Hens lay eggs if the moon is made of green cheese.
- 5) Some mice don't have bones.
- 6) If hens lay eggs then frogs are not insects.
- 7) If frogs are insects, or hens don't lay eggs, or the moon is made of green cheese, then if frogs are insects then not only is the moon made of green cheese but frogs are not insects.
- 8) None of the above.

Triangular and Square Numbers

by Brian Greene

A polygonal or figurate number is defined as any number that can be expressed as the sum of the dots composing any polygon with n sides and m dots in each side. The triangular numbers and square numbers are two types of polygonal numbers, with three and four sides respectively. They are formed in the following manner:



The first triangular number is 1, the second is 3, the third is 6, etc. The first square number is 1, the second is 4, the third is 9, etc. The question then arises: Which numbers are both triangular and square?

Now the formula for the n -th triangular number is $n(n+1)/2$ because the n -th triangular number is simply the sum of the first n natural numbers. The formula for the k -th square number is obviously k^2 . In order to find numbers that belong to the set of triangular numbers and to the set of perfect squares, we must solve:

$$(1) \quad \frac{1}{2}n(n+1) = k^2,$$

in the natural numbers. To solve this equation it is helpful to reduce it to a special form of Pell's equation

$$x^2 - dy^2 = \pm 1,$$

whose solution is well known.

Expanding (1), adding $\frac{1}{4}$ to both sides, and simplifying gives:

$$n^2 + n + \frac{1}{4} = 2k^2 + \frac{1}{4}$$

$$n + \frac{1}{2} = \sqrt{8k^2 + 1} / 2$$

$$(2) \quad 2n + 1 = \sqrt{8k^2 + 1}$$

If we let $b = 2k$ and $a = 2n + 1$, equation (2) becomes:

$$(3) \quad a^2 - 2b^2 = 1$$

Equation (3) is the desired special form of Pell's equation, with $d=2$. In general, if d is not a perfect square, Pell's equation has an infinite number of integral solutions. It was shown by Brouncker and Wallis that all solutions (x,y) can be generated from the smallest (p,q) that satisfy the equation in the following manner:

$$(x+y\sqrt{d}) = (p+q\sqrt{d})^m,$$

with m a natural number. Thus, since $a=3$, $b=2$ is the smallest solution to (3), all solutions (a,b) are equal to some $(3+2\sqrt{2})^m$. In other words, $(a+b\sqrt{2}) = (3+2\sqrt{2})^m$ where a, b are solutions to (3). It can be shown that a must be odd and b must be even to satisfy equation (3). So, $\frac{1}{2}(a-1)$, which equals n , will be an integer, and $b/2$, which equals k , will be an integer too. Thus for any natural m , we will obtain a solution to $n(n+1)/2 = k^2$.

Example: $m=3$

$$(a+b\sqrt{2}) = (3+2\sqrt{2})^3 = (99+70\sqrt{2}). \text{ Thus, } n=49 \text{ and } k=35.$$

$$49 \cdot 50 / 2 \stackrel{?}{=} 35^2 \quad 1225 \checkmark 1225$$

So the 49th triangular number is equal to the 35th square.

An interesting relation can be obtained if we notice the following:

$a^2 - 2b^2 = (a+b\sqrt{2})(a-b\sqrt{2})$ As m gets large, $(a+b\sqrt{2})$ grows without bound and $(a-b\sqrt{2})$ approaches 0.

Substituting for n and k gives:

$$n-k\sqrt{2} = (a-1)/2 - b\sqrt{2}/2.$$

Solving for k :

$k = (\sqrt{2}/2)(n+\frac{1}{2}) - (\sqrt{2}/4)(a-b\sqrt{2})$, where $(\sqrt{2}/4)(a-b\sqrt{2})$ is small and positive.

Thus, $k = [(\sqrt{2}/2)(n+\frac{1}{2})]$
if n and k are solutions to $n(n+1)/2 = k^2$.

Therefore, when n is a solution, $n(n+1)/2 = [(\sqrt{2}/2)(n+\frac{1}{2})]^2$

But $[(\sqrt{2}/2)(n+\frac{1}{2})]^2 = [(n^2+n)/2 + 1/8]$
when $(n^2+n)/2$ is an integer. Thus

$[(n^2+n)/2 + 1/8] = (n^2+n)/2$.
Thus, when n is a solution to $n(n+1)/2 = k^2$,

$$[(\sqrt{2}/2)(n+\frac{1}{2})]^2 = [\sqrt{2}/2(n+\frac{1}{2})]^2.$$

All steps in this proof are reversible, so the converse can be shown to be true. In general,

$$[x^2] \geq [x]^2.$$

So for all n :

$$[(\sqrt{2}/2)(n+\frac{1}{2})]^2 \geq [\sqrt{2}/2(n+\frac{1}{2})]^2,$$

and equality holds precisely when n is a solution to:

$$n(n+1)/2 = k^2.$$

Wallis' Product

by Steven Lev

The transcendental function $\sin x$ involves an infinite product. Helpful in deriving this formula is a knowledge of the fundamental theorem of algebra. This states that if $f(x) = a_0 + a_1x + \dots + a_nx^n$ is a polynomial of degree n and has n distinct zeros, $x_1, x_2, \dots, x_n$ then it is decomposable into linear factors:

$f(x) = a_n(x-x_1)(x-x_2)\dots(x-x_n)$. Factoring out the product $x_1x_2\dots x_n$ (assuming $x_i \neq 0$ for all i) we get:

$f(x) = c(1-x/x_1)(1-x/x_2)\dots(1-x/x_n)$, where c is a constant. Setting x equal to zero, we find that $c = a_0$.

If we now consider a function more complicated than a polynomial (a transcendental one) is a product decomposition using the zeros of $f(x)$ still attainable? This cannot always be done, for a function not crossing the x -axis (such as e^x) has no zeros; however, as Euler perceived, it is possible to factor the polynomial representation of the sine function. To simplify matters, we investigate $\sin \pi x$ instead of $\sin x$. Since $\sin \pi x$ is zero only for all integral values of x , this function has the zeros $x = 0, \pm 1, \pm 2, \dots$. From this it follows that

$$\begin{aligned}\sin \pi x &= cx(1+x/1)(1-x/1)(1+x/2)(1-x/2)(1+x/3)\dots \\ &= cx(1-x^2/1^2)(1-x^2/2^2)(1-x^2/3^2)\dots\end{aligned}$$

$$\text{Since } \lim_{x \rightarrow 0} \frac{\sin \pi x}{x} = \pi,$$

$$\text{and } \lim_{x \rightarrow 0} (1-x^2/1^2)(1-x^2/2^2)(1-x^2/3^2)\dots = 1, \text{ } c \text{ must equal } \pi,$$

so $\sin \pi x = \pi(1-x^2/1^2)(1-x^2/2^2)(1-x^2/3^2)\dots$. This infinite product converges for all x since, eventually, all the terms lie between 0 and 1.

Substituting $\frac{1}{2}$ for x , we have:

$$\sin \frac{\pi}{2} = 1 = \frac{\pi}{2} \left(1 - \frac{1}{2^2 \cdot 1^2}\right) \left(1 - \frac{1}{2^2 \cdot 2^2}\right) \left(1 - \frac{1}{2^2 \cdot 3^2}\right) \dots$$

If we express $1 - \frac{1}{2^2 \cdot n^2}$ as

$$\frac{2^2 n^2 - 1}{2^2 n^2} = \frac{(2n-1)(2n+1)}{(2n)(2n)} \quad \text{we get:}$$

$$\begin{aligned}\frac{\pi}{2} &= \lim_{n \rightarrow \infty} \frac{2}{1} \cdot \frac{2}{3} \cdot \frac{4}{3} \cdot \frac{4}{5} \dots \frac{2n}{2n-1} \cdot \frac{2n}{2n+1} \\ &= \frac{2 \cdot 2 \cdot 4 \cdot 4 \cdot 6 \cdot 6 \cdot 8 \cdot 8 \dots}{1 \cdot 3 \cdot 3 \cdot 5 \cdot 5 \cdot 7 \cdot 7 \cdot 9 \dots}\end{aligned}$$

This product was derived by the great English mathematician John Wallis(1616-1703). Many more approximations of (using continued fractions, infinite series, etc.) have been found and this is certainly a worthwhile and challenging field to explore.

Math Team

Stuyvesant's Math Team is attended by a group of exceptionally dedicated students. Each morning they gather at (about) 8 A.M. in room 517 to work on math problems and other aspects of math. Under the expert guidance of coach Mr. Barry M. Glotzer, the competing senior and junior teams condition themselves for math meets and have discussions devoted to advancing the state of the art in mathematics and improving the world in general.

Victory came to the senior team as usual, justifying the efforts of captains Greg Sorkin and Ashfaq Munshi, Irwin Jungreis, Aaron Jungreis, Mark Weiss, and alternates Steven Lev, Chris Slawinski, and Ralph Siegel. The junior team, consisting of captains Paul Feldman and Lisa Randall, Leonid Fridman, Gregg Patruno, Zachary Franco, and alternates Joel Wein, Gideon Eisenstadter, and Joel Hirschhorn, won a resounding victory, finishing 17 points ahead of the nearest competitor. In their respective divisions, Greg Sorkin, Gregg Patruno, and Zachary Franco were the top scorers.

Not only did Stuyvesant win the new New York Mathematics League competition, but had one of its students, Irwin Jungreis, achieve a perfect score.

We also did extremely well on the annual MAA exam. Irwin and Aaron Jungreis and Gregg Patruno were invited to take the Eighth United States Olympiad exam and 14 students placed on the national MAA honor roll.

The Math Team has had record attendance this year. You don't have to be a mathematician to attend. You may just want something completely different, you may want to meet new people, or you may (very unlikely) want to be entertained by the crazy antics and witty, offhand remarks of Team's leading citizens. Team will be having more lectures than in past years given by members for the remainder of this year, and we hope for increased attendance at these events. Anyone interested in giving or hearing a lecture, doing problems, or just saying hello should stop by any day during zero period.

The Extinction Probability

by Lisa Randall

If one knows the probability of a specified number of offspring being born in each generation, can one determine the extinction probability? One method of answering this question is the generating function, which incorporates the given values into an arbitrary polynomial.

Examples:

- (1) In a normal population, one might be given:
 $p(0 \text{ children}) = 1/3$, $p(1) = 1/6$, $p(2) = 1/2$. So the generating function is $f(s) = 1/3 + (1/6)s + (1/2)s^2$.
- (2) On a subway at 5:00 PM, the number of passengers might increase according to $p(i) = 1/(2^{i+1})$, $i = 0, 1, 2, \dots$
so $f(s) = 1/2 + (1/4)s + (1/8)s^2 + \dots = 1/(2-s)$ ($|s| < 2$).
- (3) The probability for a given population of bees might be:
 $p(0) = 1/3$, $p(10^6) = 2/3$ so $f(s) = 1/3 + (2/3)s^{10^6}$

Note: The generating function has no meaning yet. We will find it of use later, when it is applied in finding a way to measure the extinction probability. Since $p(i)$ represents a probability, we know that:

$$(4) \quad f(1) = \sum p(i) = 1.$$

$$(5) \quad f(0) \text{ is } p(0)$$

We also know that in general,

$$(6) \quad f(s) = \sum p(i)s^i.$$

Let $f^{(n)}(s)$ = generating function for the n -th generation probabilities and let $p(i)^n$ = probability of i individuals in the n -th generation.

We know that $f(s) = p(0) + p(1)s + p(2)s^2 + \dots$
For the second generation, $f^{(2)}(s) = \sum p(i)^2 s^i = p(0)^2 + p(1)^2 s + \dots$
One way of calculating the value of this expression would be to find the value of each of its members, eg:

$$p(0)^{(2)} = p(0) + p(1)p(0) + p(2)p(0)^2 + p(3)p(0)^3 + \dots$$

It can be seen that this method would become too messy. (For those who disagree, calculate $p(17)^{(17)}$.)

An alternative and more useful approach would be:

$$f^{(2)}(s) = p(0) + p(1)(p(0) + p(1)s + p(2)s^2 + \dots) + p(2)(f(s))^2 + p(3)(f(s))^3 + \dots$$

We see that $f^{(2)}(s) = f(f(s))$ or, in general,

$$f^{(n)}(s) = f^{(n-1)}(f(s)) = f(f^{(n-1)}(s)) = \underbrace{f(f(f \dots (s)))}_{n \text{ times}}$$

We now wish to find the extinction probability. We know that $f(0) = p(0)$ = probability of extinction by the first generation and that $p(0)^{(n)}$ = probability of extinction by the n -th generation.

$$f^{(2)}(0) = f(f(0)) = f(p(0)) = p(0)^2,$$

$$f^{(3)}(0) = f(f(f(0))) = p(0)^{(3)}.$$

It can be seen that $p(0) \quad p(0)^{(2)} \quad p(0)^{(3)} \dots 1$.

The completeness of real numbers implies that there exists an ℓ such that $\lim_{n \rightarrow \infty} p(0)^{(n)} = \ell$ (the extinction probability), where the least upper bound of $p(0)^{(n)}$ is the limit. It has already been shown that $p(0)^{(n)} = f(p(0)^{(n-1)})$.

Then f is continuous iff $\lim_{n \rightarrow \infty} f(a_n) = f(a) = f(\lim_{n \rightarrow \infty} a_n)$.

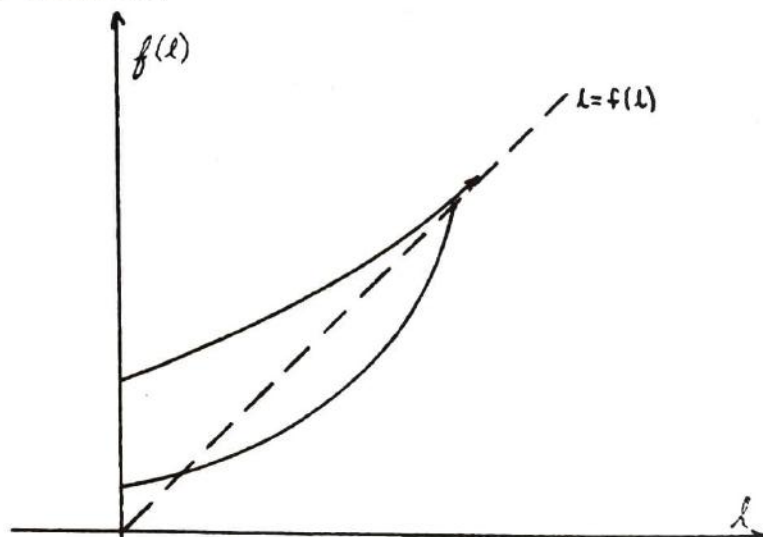
$$\lim_{n \rightarrow \infty} p(0)^{(n)} = \lim_{n \rightarrow \infty} f(p(0)^{(n-1)}) = \ell,$$

$$f(\lim_{n \rightarrow \infty} p(0)^{(n-1)}) = f(\ell)$$

so that $\ell = f(\ell)$.

This allows us to calculate the extinction probability for any set of given probabilities merely by substituting $\ell = f(\ell)$ into the equation. We refer to the previous examples:

- (1) $\ell = 1/3 + (1/6)\ell + (1/2)\ell^2$ so $\ell = 2/3$ or $\ell = 1$. We use the lower value (least upper bound) so the extinction probability is $2/3$.
- (2) $\ell = 1/(2-\ell)$ so $\ell=1$ and the population will certainly go extinct.
- (3) is left to the reader.



One might note that ℓ is the value where $y = f(x)$ intersects the line $y = x$. In order for a population to survive, this function must intersect the line $y = x$ more than once. In other words, if the function approaches 1 from underneath, then it has already crossed the line once and the population has a chance of surviving. Conversely, if it approaches from above, extinction is certain. Therefore, by finding whether the slope of the function is greater or less than one as it approaches $(1,1)$, the fate of a given population can be ascertained.

Cardinality of Functions

by Leonid Fridman

It is easy to see that not every curve on the plane is a graph of a polynomial relation with finitely many terms. For example, the graph of $y = \sin(x)$ is not a graph of such a relation. However, we will show here by means of set theory that most of the curves on a plane are not the graphs of polynomials of finitely many terms with real coefficients.

Let us introduce some notation:

A - the power (number of elements) of a countable set, such as the set of integers. (A more popular but less printable notation for it is $\aleph_0$)

C - (also called continuum) - the power of the set of real numbers, or any other set equipollent to it. ($\aleph_1$ in other notation.)

In our proof we will use three lemmas:

Lemma 1: $A \leq C$

Lemma 2: $C^n = C$ for $n = 1, 2, 3, \dots$

Lemma 3: For every set M, the set $U(M)$ of all its subsets has a greater cardinality (number of elements) than M.

Theorem: The power P of the set of all curves given by polynomial formulas $f(x,y) = 0$ with real coefficients and of finite length is less than the power of the set of other curves on the plane.

Proof: First we will prove $P = C$. We can see that P is equal to the power of the set of polynomial functions $f(x,y)$, since we have a one-to-one mapping between relations $f(x,y)=0$ and functions $f(x,y)$.

Let us define the height of a polynomial to be the largest of the sums of the powers of x and y in the terms of the polynomial. For example, the height of $3x^5y^4 + 5x^9y^2 + 17x^3 + y$ is $9 + 2 = 11$. The general polynomial of height 3 would be:

$b_1x^3 + b_2x^2 + b_3x + b_4 + b_5x^2y + b_6xy + b_7y + b_8xy^2 + b_9y^2 + b_{10}y^3$.
And the general polynomial of height h would be:

$$\sum_{i=0}^h k_i x^i \cdot \sum_{j=0}^{h-i} k_j x^j$$

Thus, every polynomial of height n can be represented by an ordered sequence $(b_1, b_2, b_3, \dots, b_{\frac{1}{2}(n+1)(n+2)})$ where b_t is the coefficient of the t-th term in our general representation of a polynomial of height n. For example, $(2, 0, 0, 0, 3, 0)$ corresponds to $2x^2 + 3y$. So, there are as many polynomials of height n as there are permutations of $\frac{1}{2}(n+1)(n+2)$ real numbers.

Since the power of the set of real numbers is C , the power of the set of polynomial functions of height n is $C^{n+1}(n+2)$. But, according to Lemma 2, $C^{n+1}(n+2) = C$. So, there are continuum-many polynomials of height n , so the power of the set of all polynomials is $C \cdot A$. But $C \cdot A$, according to lemma 1, and thus (using also Lemma 2) $C \cdot C \cdot A = C \cdot C = C$, so $C \cdot A = C$. Thus, the power of the set of all polynomial functions is C , and $P = C$.

Now we will show that the power of the set of all curves on the plane, which we will denote by S , is equal to 2^C . Suppose we were tracing a curve and stopped at some point. Then there would be continuum-many directions we could continue our curve. Since there are C points on a general curve, there are at least C^C ways we could draw it. So $C^C \leq S$. Since $C \geq 2$, we get $2^C \leq C^C \leq S$. But there are C points on the plane, since each point can be represented as an ordered pair of real numbers, and $C \cdot C = C^2 = C$ according to Lemma 2. Thus, the power of the set of all subsets of a plane is 2^C , since every other point either belongs or doesn't belong to a given subset. Now any curve on the plane is just one subset of the plane, so $S \leq 2^C$. Hence $2^C \leq S \leq 2^C$ and thus, $S = 2^C$. Since the power of the set of all subsets of a set is greater than the power of the set itself, (according to Lemma 3), we get $2^C > C$. Hence, $S = 2^C > C = P$, and we conclude that $S > P$, i.e. the power of the set of all curves on the plane is greater than the power of the set of all curves given by the polynomial formulas with real coefficients and of finite length.

Finally, we will prove that $S = P$, which will finish the proof of the theorem since $S = P$ is equal to the power of the set of non-polynomial curves on the plane by the definitions of S and P . Suppose $S = P$. Then $S = 2^P = 2^C = 2C$, since $S = 2^C$, and $P = C$. But $C = 2C = C \cdot C = C$ so $2C = C = 2^C = C$. Since we also know $2^C > C$ (from Lemma 3) we obtain a contradiction. So we must indeed have $S = P = P$. We thus conclude that the power of the set of non-polynomial curves is greater than the power of the set of polynomial curves. In fact, the power of the set of non-polynomial curves is, in a sense, "much" greater than P , for $S = P = C \cdot P$ follows directly from the fact that $C \cdot P = C \cdot C = C = P$.

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Stirling's Approximation

by Irwin Jungreis

When dealing with the function $f(x) = x!$, it is often useful to have an approximation which is good for large values of x . The one most commonly used, Stirling's approximation, states that:

$$x! \sim x^x e^{-x} \sqrt{2\pi x}$$

There are many proofs of this, some using the gamma function, most using infinite series or products, some involving complex variables, and many of which employ a good deal of mathematical "fudge". The following proof, developed by Irwin Jungreis (1963-) uses none of these, is completely rigorous, and never goes beyond the level of elementary calculus.

First, let us examine the curve $y = \ln(x)$ for $1 \leq x \leq 2$. The area $K = \int_1^2 \ln(x) dx$. This can be integrated by parts yielding

(1) $K = 2\ln(2) - 1$. Since $y = \ln(x)$ is always concave down, a trapezoidal approximation will be an under-approximation. Take trapezoids of width $1/n$.

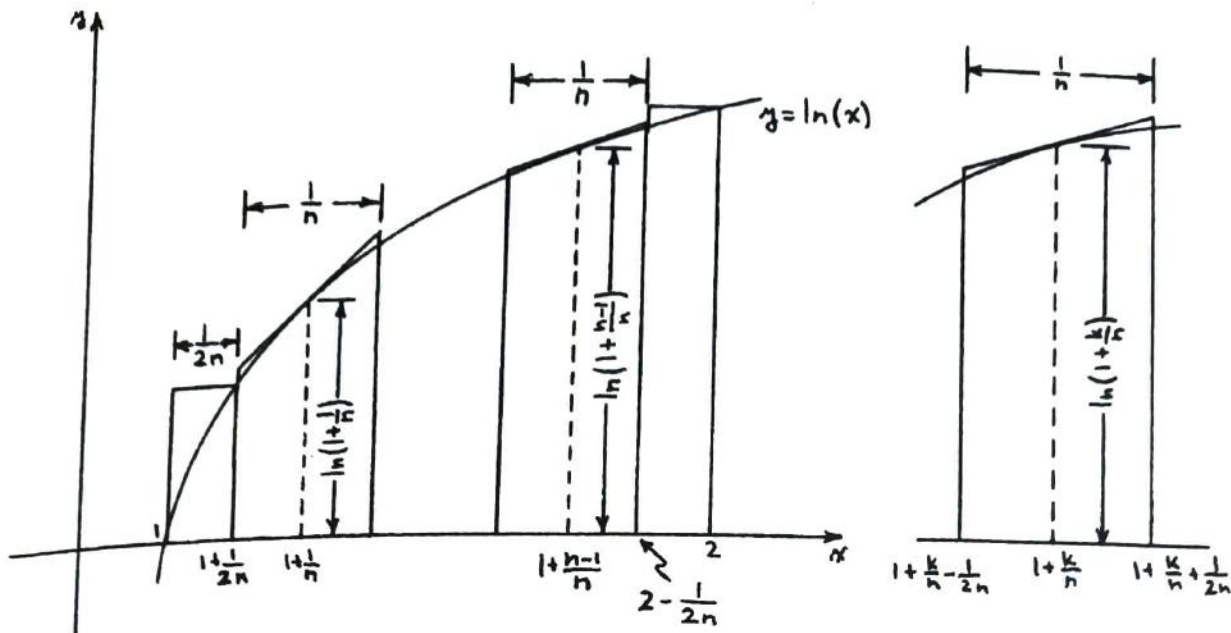
$$\begin{aligned} K &> \frac{1}{n} \left[\frac{1}{2} \ln(1) + \ln\left(1 + \frac{1}{n}\right) + \ln\left(1 + \frac{2}{n}\right) + \dots + \ln\left(1 + \frac{n-1}{n}\right) + \frac{1}{2} \ln\left(1 + \frac{n}{n}\right) \right] \\ &= \frac{1}{n} \left[0 + \ln\left(1 + \frac{1}{n}\right) + \ln\left(1 + \frac{2}{n}\right) + \dots + \ln\left(1 + \frac{n-1}{n}\right) + \ln\left(1 + \frac{n}{n}\right) - \frac{1}{2} \ln\left(1 + \frac{n}{n}\right) \right] \\ &= \frac{1}{n} \left[\ln\left(\frac{n+1}{n}\right) + \ln\left(\frac{n+2}{n}\right) + \dots + \ln\left(\frac{2n-1}{n}\right) + \ln\left(\frac{2n}{n}\right) - \frac{1}{2} \ln(2) \right] \\ &= \frac{1}{n} \ln \frac{(n+1)(n+2)\dots(2n)}{n^n \sqrt{2}} = \frac{1}{n} \ln \frac{(2n)!}{n! n^n \sqrt{2}} \end{aligned}$$

$$(2) \quad K > \frac{1}{n} \ln \frac{(2n)!}{n! n^n \sqrt{2}}$$

We may get an over-approximation as follows. For $k=1, 2, 3, \dots, n-1$ we take the tangent to the curve at $x = 1 + \frac{k}{n}$ as one side of a trapezoid (refer to diagram), the other three sides being the x -axis and the lines $x = 1 + \frac{k}{n} + \frac{1}{2n}$ and $x = 1 + \frac{k}{n} - \frac{1}{2n}$. The average of the bases will be $\ln(1 + \frac{k}{n})$, and the height will be $\frac{1}{n}$, so the area will be $\frac{1}{n} \ln(1 + \frac{k}{n})$. If we add up all such areas we will get an overestimate of the area under the curve in the interval $1 + \frac{1}{2n} \leq x \leq 2 - \frac{1}{2n}$. The extra pieces for $1 \leq x < 1 + \frac{1}{2n}$ and $2 - \frac{1}{2n} < x \leq 2$ can be approximated by rectangles of width $\frac{1}{2n}$ and heights $\ln(1 + \frac{1}{2n})$ and $\ln(2)$ respectively.

$$K < \frac{1}{n} \ln\left(1 + \frac{1}{n}\right) + \frac{1}{n} \ln\left(1 + \frac{2}{n}\right) + \dots + \frac{1}{n} \ln\left(1 + \frac{n-1}{n}\right) + \frac{1}{2n} \ln\left(1 + \frac{1}{2n}\right) + \frac{1}{2n} \ln(2)$$

by algebraic manipulation we obtain:



(3) $K < \frac{1}{n} \ln \frac{(2n)!}{n! n^n \sqrt{2}} \sqrt{1 + \frac{1}{2n}}$ So, from (1), (2), and (3):

$\frac{1}{n} \ln \frac{(2n)!}{n! n^n \sqrt{2}} \sqrt{1 + \frac{1}{2n}} > 2 \ln(2) - 1 > \frac{1}{n} \ln \frac{(2n)!}{n! n^n \sqrt{2}}$ and we obtain

$\frac{(2n)! \sqrt{1 + 1/2n}}{n^n n! \sqrt{2}} > 2^{2n} e^{-n} > \frac{(2n)! \sqrt{1 + 0/2n}}{n^n n! \sqrt{2}}$

so $2^{2n} e^{-n} = \frac{(2n)!}{n^n n! \sqrt{2}} \sqrt{1 + k/2n}$ where k is a function of n and $0 < k < 1$

for all n . Isolating $\frac{(2n)!}{n!}$ we get

(4) $\frac{(2n)!}{n!} = \frac{2^{2n} e^{-n} n^n \sqrt{2}}{\sqrt{1 + k/2n}}$

This is true for all positive integers n , not just for large ones.

Now, let us look at

$\int_0^{\pi/2} \cos^m x dx$ for $m \in \mathbb{N}^+$

This can be evaluated by parts.

$\cos^{m-1} x = v \quad \cos x dx = du$

$-(m-1) \cos^{m-2} x \sin x dx = dv \quad \sin x = u$

$\int_0^{\pi/2} \cos^m x dx = \sin x \cos^{m-1} x \Big|_0^{\pi/2} + (m-1) \int_0^{\pi/2} \cos^{m-2} x \sin^2 x dx$

this can be simplified using the pythagorean identity

$\sin^2 x = 1 - \cos^2 x$

$\int_0^{\pi/2} \cos^m x dx = (m-1) \int_0^{\pi/2} \cos^{m-2} x dx - (m-1) \int_0^{\pi/2} \cos^m x dx$

$m \int_0^{\pi/2} \cos^m x dx = (m-1) \int_0^{\pi/2} \cos^{m-2} x dx$

$\int_0^{\pi/2} \cos^m x \, dx = \frac{m-1}{m} \int_0^{\pi/2} \cos^{m-2} x \, dx$
 similarly $\int_0^{\pi/2} \cos^{m-2} x \, dx = \frac{m-3}{m-2} \int_0^{\pi/2} \cos^{m-4} x \, dx$. This process can
 be repeated until we are left with either $\int_0^{\pi/2} \cos x \, dx$ or
 $\int_0^{\pi/2} dx$, depending on whether m is odd or even.

for $m = 2n$

$$\begin{aligned}
 (5) \quad \int_0^{\pi/2} \cos^{2n} x \, dx &= \frac{2n-1}{2n} \int_0^{\pi/2} \cos^{2n-2} x \, dx = \frac{2n-1}{2n} \cdot \frac{2n-3}{2n-2} \cdot \dots \\
 &\cdot \frac{1}{2} \int_0^{\pi/2} dx = \frac{2n-1}{2n} \cdot \frac{2n-3}{2n-2} \cdot \dots \cdot \frac{1}{2} \cdot \frac{\pi}{2}
 \end{aligned}$$

for $m = 2n+1$

$$\begin{aligned}
 (6) \quad \int_0^{\pi/2} \cos^{2n+1} x \, dx &= \frac{2n}{2n+1} \int_0^{\pi/2} \cos^{2n-1} x \, dx = \\
 &\frac{2n}{2n+1} \cdot \frac{2n-2}{2n-1} \cdot \dots \cdot \frac{2}{3} \int_0^{\pi/2} \cos x \, dx = \frac{2n}{2n+1} \cdot \frac{2n-2}{2n-1} \cdot \dots \cdot \frac{2}{3}
 \end{aligned}$$

For $0 \leq x \leq \pi/2$, $0 \leq \cos x \leq 1$ and $0 \leq \cos^k x \leq 1$; therefore, for all x
 in that interval $\cos^{k+1} x \leq \cos^k x$ so the area under $y = \cos^{k+1} x$
 must be less than the area under $y = \cos^k x$.

$$\begin{aligned}
 \int_0^{\pi/2} \cos^{k+1} x \, dx &< \int_0^{\pi/2} \cos^k x \, dx \\
 \int_0^{\pi/2} \cos^{2n-1} x \, dx &> \int_0^{\pi/2} \cos^{2n} x \, dx > \int_0^{\pi/2} \cos^{2n+1} x \, dx \\
 \frac{2n-2}{2n-1} \cdot \frac{2n-4}{2n-3} \cdot \dots \cdot \frac{2}{3} &> \frac{\pi}{2} \cdot \frac{2n-1}{2n} \cdot \frac{2n-3}{2n-2} \cdot \dots \cdot \frac{1}{2} > \frac{2n}{2n+1} \cdot \frac{2n-2}{2n-1} \cdot \dots \cdot \frac{2}{3}
 \end{aligned}$$

$$\frac{(2n)(2n-2)^2(2n-4)^2 \dots 2^2}{(2n-1)^2(2n-3)^2 \dots 3^2 \cdot 1^2} > \frac{\pi}{2} > \frac{(2n)^2(2n-2)^2 \dots 2^2}{(2n+1)(2n-1)^2(2n-3)^2 \dots 3^2 \cdot 1^2}$$

If we multiply numerator and denominator of the expression on the
 left by $(2n)^3(2n-2)^2(2n-4)^2 \dots 2^2$, we obtain:

$$\begin{aligned}
 \frac{(2n)^4(2n-2)^4(2n-4)^4 \dots 2^4}{(2n)(2n)^2(2n-1)^2(2n-2)^2 \dots 3^2 \cdot 2^2 \cdot 1^2} &= \frac{2^{4n}(n!)^4(n-1)^4 \dots (2)^4}{(2n)(2n)^2(2n-1)^2 \dots 3^2 \cdot 2^2 \cdot 1^2} \\
 = \frac{2^{4n}(n!)^4}{2n(2n)!^2} &\quad \text{The expression on the right is this times } 2n/(2n+1). \\
 \frac{2^{4n}(n!)^4}{4(2n)!^2(2n)} &> \frac{\pi}{2} > \frac{2^{4n}(n!)^4}{(2n)!^2(2n+1)}
 \end{aligned}$$

$$\frac{\pi}{2} = \frac{2^{4n}(n!)^4}{(2n)!^2(2n+j)} \quad \text{where } j \text{ is a function of } n \text{ and } 0 < j < 1 \text{ for all } n.$$

$$(n!)^2 = \frac{\pi}{2}(2n+j)2^{-4n} \left[\frac{(2n)!}{n!} \right]^2 \text{ but from (4) we know that}$$

$$\frac{(2n)!}{n!} = \frac{2^{2n} e^{-n} n^n \sqrt{2}}{\sqrt{1+k/n}}$$

$$n!^2 = \frac{\pi}{2}(2n+j)2^{-4n} \left[\frac{2^{2n} e^{-n} n^n \sqrt{2}}{\sqrt{1+k/2n}} \right]^2 = \frac{\pi(2n+j)2^{-4n} 2^{4n} e^{-2n} n^{2n} (2)}{2(1+k/2n)}$$

cancelling the powers of 2 and multiplying numerator and denominator by 2n we get:

$$n^{2n} e^{-2n} (2\pi n) \frac{2n+j}{2n+k}$$

$$n! = n^n e^{-n} \sqrt{2\pi n} \sqrt{\frac{2n+j}{2n+k}}$$

Since $0 < j < 1$ and $0 < k < 1$, the lower limit occurs when $j = 0$, $k = 1$ and the upper one occurs when $j = 1$ and $k = 0$.

$$n^n e^{-n} \sqrt{2\pi n(2n)/(2n+1)} < n! < n^n e^{-n} \sqrt{2\pi n(2n+1)/(2n)}$$

$$n^n e^{-n} \sqrt{2\pi n(\sqrt{1-1/(2n+1)})} < n! < n^n e^{-n} \sqrt{2\pi n(\sqrt{1+1/(2n)})}$$

Again, this holds for all $n \in \mathbb{N}^+$, not just large n . If we wish a neater expression and are willing to restrict our attention to large values of n , we notice that for this case,

$$1 - 1/(2n+1) \sim 1 + 1/(2n) \sim 1$$

therefore
$$n! \sim n^n e^{-n} \sqrt{2\pi n}.$$

PROBLEMS

Find the radius-length of the circle whose area is tripled when the length of its radius is increased by 2.

Find all ordered pairs of real numbers (x, y) for which $\frac{y \log_2 x}{y^2 - 13} = -\frac{1}{4}$ and $x = 2^{3y}$.

For what value of x does the continued fraction

$$x + \frac{1}{x + \frac{1}{x + \frac{1}{x + \dots}}}$$

have the value 3, if the dots indicate an infinite repetition of the indicated pattern?

n-th Powers of Integers

by Ashfaq Munshi

If you write down the sequence of 4-th powers of the positive integers, then write the differences below these numbers, and repeat this process five times, you get:

0	1	16	81	256	625	1296	...
1	15	65	175	369	671	...	
	14	50	110	194	302	...	
		36	60	84	108	...	
			24	24	24	...	
			0	0	...		

Notice that each term in the second to the last row is $4! = 4 \cdot 3 \cdot 2 \cdot 1 = 24$. You might ask whether you'll get $3! = 3 \cdot 2 \cdot 1 = 6$ if you do the same with the third powers. In fact, this will be true for n-th powers of positive integers; that is to say that if you listed the n-th powers and constructed a table as we did earlier, the next to the last line would be a sequence of n!.

A possible approach to the proof might be to see how the array with which we started leads to compact formulas. Observe the following computations made in forming the table:

$$\begin{array}{ll} \text{second} & (256 - 81) - (81 - 65) = (1)256 - (2)81 + (1)65 = 110 \\ \text{differences} & (81 - 16) - (16 - 1) = (1)81 - (2)16 + (1)1 = 50 \\ & (16 - 1) - (1 - 0) = (1)16 - (2)1 + (1)0 = 14 \end{array}$$

$$\begin{array}{ll} \text{third} & (1)256 - (3)81 + (3)16 - (1)1 = 110 - 50 = 60 \\ \text{differences} & (1)81 - (3)16 + (3)1 - (1)0 = 50 - 14 = 36 \end{array}$$

$$\begin{array}{ll} \text{fourth} & (1)256 - (4)81 + (6)16 - (4)1 + (1)0 = 60 - 36 = 24 \\ \text{differences} & \end{array}$$

$$\begin{array}{ll} \text{fifth} & (1)625 - (5)256 + (10)81 - (10)16 + (5)1 - (1)0 = 0 \\ \text{differences} & \end{array}$$

Notice that the binomial coefficients appear in the table quite naturally. The computations for the last two rows can be written compactly as:

$$\begin{array}{l} \text{and} \quad \sum_{j=0}^4 (-1)^j \binom{4}{j} (4-j)^4 = \sum_{j=0}^4 (-1)^{4-j} \binom{4}{j} j^4 = 4! \\ \quad \sum_{j=0}^5 (-1)^j \binom{5}{j} (5-j)^5 = \sum_{j=0}^5 (-1)^{5-j} \binom{5}{j} j^5 = 0, \end{array}$$

where $\binom{n}{j} = \frac{n!}{(n-j)! j!}$.

It would then seem that in general,

$$\sum_{j=0}^n (-1)^{n-j} \binom{n}{j} j^k = \begin{cases} 0 & \text{if } 0 \leq j < n \\ n! & \text{if } j=n \end{cases}$$

We will now prove this conjecture, known as Euler's formula, by induction. First, we need the following facts:

(1) $\binom{n+1}{j} = \binom{n}{j} + \binom{n}{j-1}$, and

(2) $\binom{n}{j} = 0$ for $0 \leq n < j$ or $j < 0$.

Now let $f(n, k) = \sum_{j=0}^n (-1)^{n-j} \binom{n}{j} j^k$. Then the inductive proof is as follows:

$$\begin{aligned} f(n+1, k) &= \sum_{j=0}^{n+1} (-1)^{n+1-j} \binom{n+1}{j} j^k \\ &= \sum_{j=0}^{n+1} (-1)^{n+1-j} \left[\binom{n}{j} + \binom{n}{j-1} \right] j^k \end{aligned}$$

where we have used (1) to expand $\binom{n+1}{j}$

since $(-1)^{n+1-j} = (-1)(-1)^{n-j} = -(-1)^{n-j}$

we have, $f(n+1, k) = - \sum_{j=0}^{n+1} (-1)^{n-j} \binom{n}{j} j^k + \sum_{j=0}^n (-1)^{n-j} \binom{n}{j} (j+1)^k$

where the last summation has been changed to facilitate computation. Now, the first summation is simply $-f(n, k)$.

We then expand $(j+1)^k$. This is simply $\sum_{r=0}^k \binom{k}{r} j^r$. Plugging

into the second summation gives

$$\sum_{j=0}^n (-1)^{n-j} \binom{n}{j} (j+1)^k = \sum_{r=0}^k \binom{k}{r} \sum_{j=0}^n (-1)^{n-j} \binom{n}{j} j^r$$

Note once again that

$$\sum_{j=0}^n (-1)^{n-j} \binom{n}{j} j^r = f(n, r). \text{ Thus,}$$

$$f(n+1, k) = -f(n, k) + \sum_{r=0}^k \binom{k}{r} f(n, r), \quad 0 \leq k < n$$

or $f(n+1, k) = \sum_{r=0}^{k-1} \binom{k}{r} f(n, r)$

Therefore, if $f(n, k) = 0$ then $f(n+1, k) = 0$, for $0 \leq k < n$. Since $f(1, k) = 0$ for $0 \leq k < 1$, $f(2, k) = 0$, etc. and the proof is complete.

Let $g(n) = f(n, n)$. We want to prove that $g(n) = n!$. Once again we use induction:

$$g(n+1) = \sum_{j=0}^{n+1} (-1)^{n+1-j} \binom{n+1}{j} j^{n+1} = (n+1) \sum_{j=1}^{n+1} (-1)^{n+1-j} \binom{n}{j-1} j^n$$

Now, $j \binom{n+1}{j} = (n+1) \binom{n}{j-1}$. Changing the variable index j to $j+1$ we have

$$g(n+1) = (n+1) \sum_{j=0}^n (-1)^{n-j} \binom{n}{j} (j+1)^n. \text{ Using the binomial theorem}$$

and expanding $(j+1)^n$ gives:

$$\begin{aligned} g(n+1) &= (n+1) \sum_{k=0}^n \binom{n}{k} \sum_{j=0}^n (-1)^{n-j} \binom{n}{j} j^k \\ &= (n+1) \sum_{k=0}^n \binom{n}{k} f(n,k) = (n+1)g(n), \text{ where we have the} \end{aligned}$$

recurrence relation $g(n+1) = (n+1)g(n)$. We know that $g(0) = 1$, then $g(1) = 1$, $g(2) = 2$, $g(3) = 6$ and more generally $g(n) = n!$ which is what we wanted to prove.

continued from page 10

$$5k/10 = m/4 + p/3 + m/8 = 3m/8 + p/3.$$

$$p = k/5 + m/4 + p/3, \quad 2p/3 = k/5 + m/4, \text{ so } 5k/10 = 3m/8 + (1/5)(k/2) + (1/4)(m/2), \quad 4k/10 = 4m/8, \quad k = 5m/4.$$

$$5m/8 = k/2 = 3m/8 + p/3, \quad 2m/8 = p/3, \quad 4k/5 = m = 4p/3, \quad k = 5p/3.$$

$$q = k/5 + m/4 = k/5 + (1/4)(4k/5) = 2k/5, \quad r = k/5.$$

Thus, $r + q + p + m + k = 15k/5 = 3k$, and since this represents a sure event, $k = 1/3$.

Now, if x is the probability of hitting n , we can express the event of hitting $n-1$ and not hitting n in two ways:

- 1) The chance of hitting $n-1$ and then rolling a 2, 3, 4, 5, or 6, with probability $5x/6$; or
 - 2) The chance of not hitting n , $1-x$, times k , which equals $1/3$.
- Thus, $5x/6 = (1/3)(1-x)$, $7x/6 = 1/3$, $x = 2/7$.

This agrees with an instinctive guess one might have, since the rolls should average about $7/2$, one out of every $7/2$ or two out of every seven would be hit.

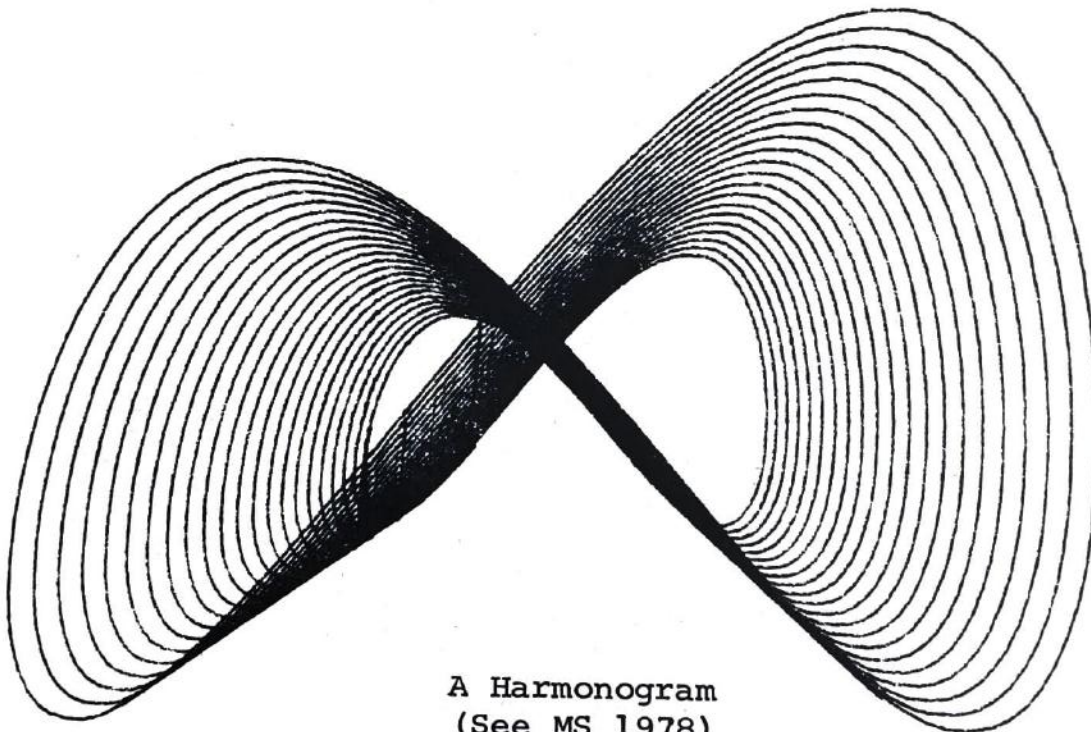
Note from the editors:

We would like to express our sincere and heartfelt thanks to the Trenk family for allowing us the use of their home while putting together this magazine.

continued from page 20

Since B is the only one who could have encoded his name with f_B^{-1} (he is the only person who knows it), the message must indeed have been from him. While the computer program method provides one set of trap-door one-way functions, other methods exist and include polynomial functions and raising to powers in a modulus.

It is interesting to note that while the public-key cryptosystem appears unheralded, it was preceded by a trend of less reliance on the secrecy of the system and more on the key. Tied in with this and the developments in computers is a shift back to the determination of the quality of a cryptosystem by mathematical means rather than the subjective quality of how long it takes a team of experts to break it. And a final word for those interested in cryptography: while the experts are best at producing and cracking good cryptosystems, most innovation in the field has come from amateurs.



A Harmonogram
(See MS 1978)

Compliments of the

Parents Association

NEW YORK UNIVERSITY
DEPARTMENT OF MATHEMATICS

PRESENTS:

....SOME BRAIN TEASERS TO TEST YOUR METTLE. A SPLENDID AWARD
WILL BE MADE FOR THE BEST SOLUTIONS. GOOD LUCK!

1. Let $P_1, P_2, \dots, P_n$ be the consecutive vertices of a regular n -gon inscribed in a circle of radius 1. Evaluate

$$A(n) = \prod_{\substack{k=\text{odd} \\ 1 \leq k < n}} \overline{P_k P_n} \quad \text{and} \quad B(n) = \prod_{\substack{k=\text{even} \\ 1 \leq k < n}} \overline{P_k P_n}$$

2. Prove

$$\det \begin{vmatrix} 1 & x_1 & x_1^2 & \dots & x_1^{n-2} & x_1^{n+1} \\ 1 & x_2 & x_2^2 & \dots & x_2^{n-2} & x_2^{n+1} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 1 & x_n & x_n^2 & \dots & x_n^{n-2} & x_n^{n+1} \end{vmatrix}$$

$$= \det \begin{vmatrix} 1 & x_1 & x_1^2 & \dots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \dots & x_2^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \dots & x_n^{n-1} \end{vmatrix} \left(\sum_{i=1}^n x_i^2 + \sum_{1 \leq i < j \leq n} x_i x_j \right)$$

3. Show that one of $\pi + e$ and $\pi \cdot e$ is transcendental.
4. Let C be a convex polyhedron. Show that C has a face F on which C can rest without tipping over.

Send your solutions to: DEPARTMENT OF MATHEMATICS c/o M. Doyle
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DEADLINE: July 1, 1979